

Arnav Bansal

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EDUCATION

University of Illinois Urbana-Champaign

Champaign-Urbana, IL

Bachelor of Science in Computer Science | GPA: 4.0/4.0

Aug. 2025 – May 2029

Relevant Coursework: Data Structures, Algorithms, Computer Architecture, Probability & Statistics, Data Science

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

Distributed Protocols Research Group, UIUC

Urbana, IL

- Built a high-throughput benchmarking framework in **Go** to **evaluate distributed LLM inference systems** across **seven workloads and ~250k requests**, including chat, RAG, long-context, code, and agentic traces, measuring **TTFT, latency, token throughput, and KV-cache behavior** under controlled prefix reuse and bursty traffic.
- Developed **agentrecap**, a tool to analyze local coding agent (Claude Code, Codex, Cursor, OpenCode etc.) sessions, studying cost, tool use, latency, token/cache efficiency, failures, and subagent behavior.

Software Engineering Intern

Jun. 2026 – Aug. 2026

Open Science Stack

Delhi, India

- Built a **deterministic verification pipeline** for LLM-generated laboratory code, detecting **55+ classes of errors** before physical execution.
- Designed a two-stage validation system combining **static and dynamic analysis**, using **AST parsing** and language-server diagnostics to traverse every reachable branch before execution in a stateful simulator.

Software Engineering Intern

Mar. 2024 – Oct. 2024

InfoEdge India Pvt. Ltd.

Delhi, India

- Built a server-rendered career counseling platform with **React and Next.js**, including a context-aware AI assistant backed by persistent user history.
- Designed a **semantic summarization algorithm** that reduced prompt token usage by **80%** while preserving response quality and personalization.

PROJECTS

Alfie | TypeScript, Hono, Cloudflare, PostgreSQL, Redis

2026 – Present

- Built a **text-based personal AI agent** with memory, web search, browser use, and durable scheduled tasks.
- Designed an actor-based main/worker agent architecture with asynchronous message-passing using **Cloudflare Durable Objects** to keep conversations responsive while coordinating multiple **long-running tasks**.
- Implemented a **sandboxed, extensible plugin system** using **QuickJS**, enabling users to connect third-party services and AI agents to **create and modify integrations at runtime** on demand.

Railyard | Rust, Pingora, Podman

2026 – Present

- Building a self-hosted PaaS in **Rust**, engineering a background **systemd daemon** and local CLI to manage VPS infrastructure and container lifecycles.
- Developed a **Pingora-based reverse proxy**, with **automatic TLS/SSL termination** and service routing.
- Architected a **Podman container orchestration engine** supporting **OCI runtimes**, persistent volumes, health-gated zero-downtime rollouts, automatic rollbacks, and private overlay networking.
- Built a secure, declarative **control plane** with **Ed25519-signed requests**, project-scoped access, single-use invitations, and a **typed manifest** for **deterministic updates by human developers and AI agents**.

Cyclr | React Native, Bun/Elysia, PostgreSQL, ESP32

2025

- Built an end-to-end **bike-sharing platform**, including a React Native app, Bun/Elysia API, PostgreSQL backend, and custom IoT smart lock built with an **ESP32**.
- Developed **ESP32 firmware** to handle BLE/GATT service advertising and communication, and **HMAC-SHA256 challenge-response authentication** to securely control physical locks, preventing replay attacks.
- Won at UIUC's Technology Entrepreneur Center's Idea Fair.

TECHNICAL SKILLS

Languages: Rust, Go, C/C++, TypeScript/JavaScript, Python, SQL, Dart, Kotlin, Bash

Frameworks & Libraries: React, Next.js, Hono, Elysia, React Native, Flutter, Pandas, NumPy, Durable Objects

Systems & Tools: Linux, Git, Docker, Podman, Kubernetes, BuildKit, Pingora, PostgreSQL, Redis, Workflows, IoT